FORM TP 2016157

TEST CODE **02168020**

MAY/JUNE 2016

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

BUILDING AND MECHANICAL ENGINEERING DRAWING

OPTION A – MECHANICAL ENGINEERING DRAWING

UNIT 1 – Paper 02

2 hours 30 minutes

09 MAY 2016 (a.m.)

GENERAL INFORMATION

1. This paper consists of SIX questions. EACH question is worth 20 marks.
2. Answer ALL questions.
3. Silent, non-programmable calculators may be used for this examination.
4. For this examination, each candidate should have:

   Two sheets of drawing paper (both sides may be used)
   Drawing board and T-square
   Drawing instruments
   Data sheet B.S.4500 (both hole and shaft basis), provided as an insert

INSTRUCTIONS TO CANDIDATES

1. All dimensions given are in millimetres unless otherwise stated.
2. Where scales are not stated, the full size should be applied.
3. All geometrical construction lines MUST be visible on all drawings.
4. When first-angle or third-angle is not specified, the choice of projection is left to the candidate's discretion, in which case the type of projection used MUST be clearly stated.
5. The candidate should use his/her own judgement to supply any dimensions or other details not directly shown on the drawings.
6. The number of each question MUST be written next to the solution.
7. Each candidate MUST enter his/her school code and registration number in the appropriate space at the bottom of the drawing paper.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

MODULE 1 — GEOMETRY 1

Answer ALL questions from this section.

1. Figure 1 shows the framework for a point, P, which moves so that its distance from the circumference of two circles, as well as from their centres O_1 and O_2, is always in the ratio 2:3. The radii of the two circles are 20 mm and 15 mm respectively.

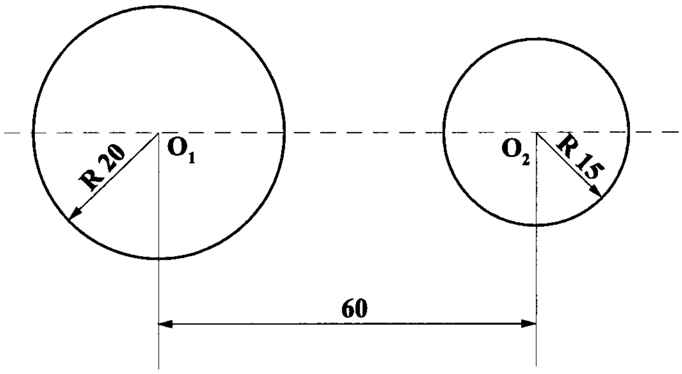

Figure 1

(a) Plot the locus of the point P. **[5 marks]**

(b) State the name of the locus. **[1 mark]**

(c) Sketch, freehand, an ellipse with minor and major axes of 50 mm and 100 mm respectively. On the sketch, label the following elliptical terms:

(i) Chord

(ii) Ordinate **[3 marks]**

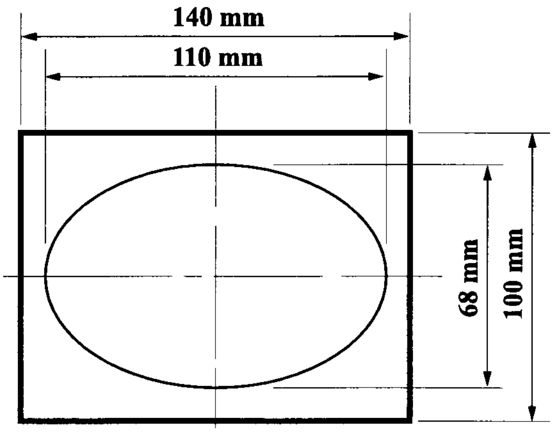

Figure 2

(d) Figure 2 shows the loudspeaker grill of a car radio. The grill is rectangular with an elliptical hole.

(i) Draw the grill, full size, clearly showing the construction of the ellipse. **[9 marks]**

(ii) Find the foci of the ellipse drawn in (i). **[2 marks]**

Total 20 marks

2. Figure 3 shows a symmetrical shield.

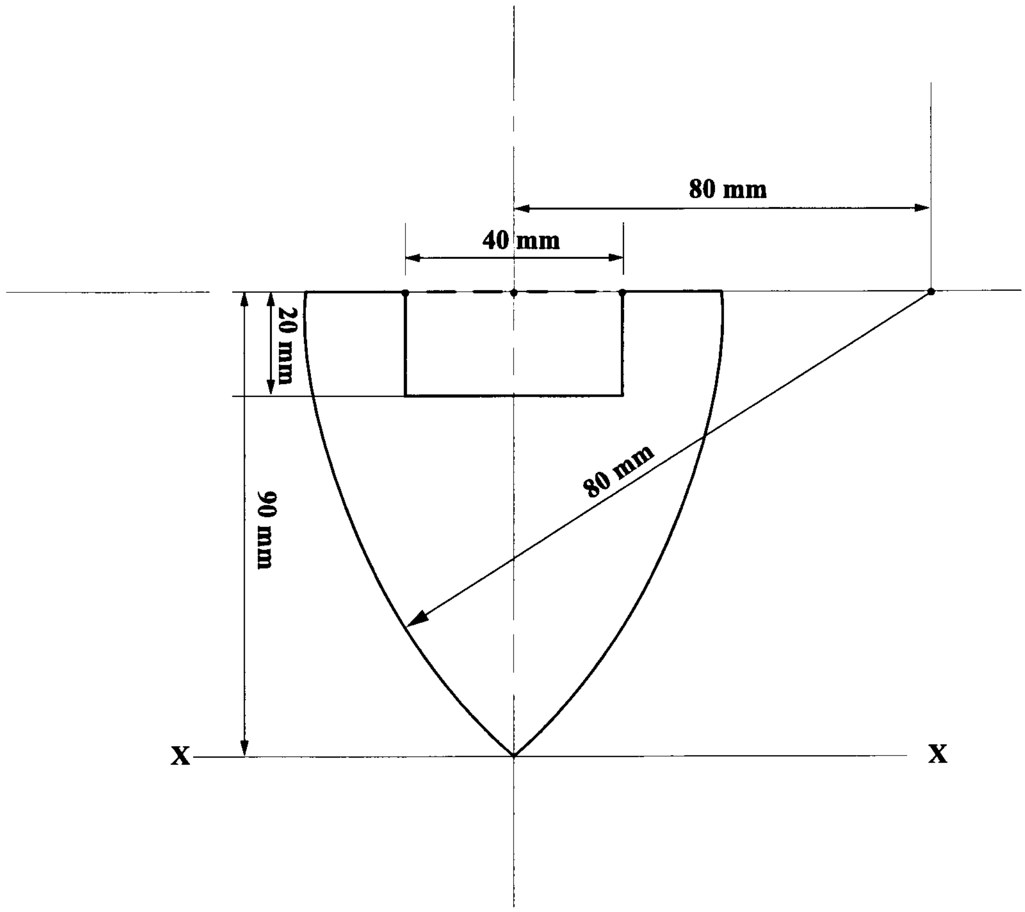

Figure 3

Determine the position of the centroid from XX, using graphical integration (use a distance to the pole of 50 mm).

Hence, determine the second moment of area. **[20 marks]**

Total 20 marks

SECTION B

MODULE 2 — GEOMETRY 2

Answer ALL questions from this section.

3. Figure 4 shows two views of two spheres intersecting. The hidden details are not shown.

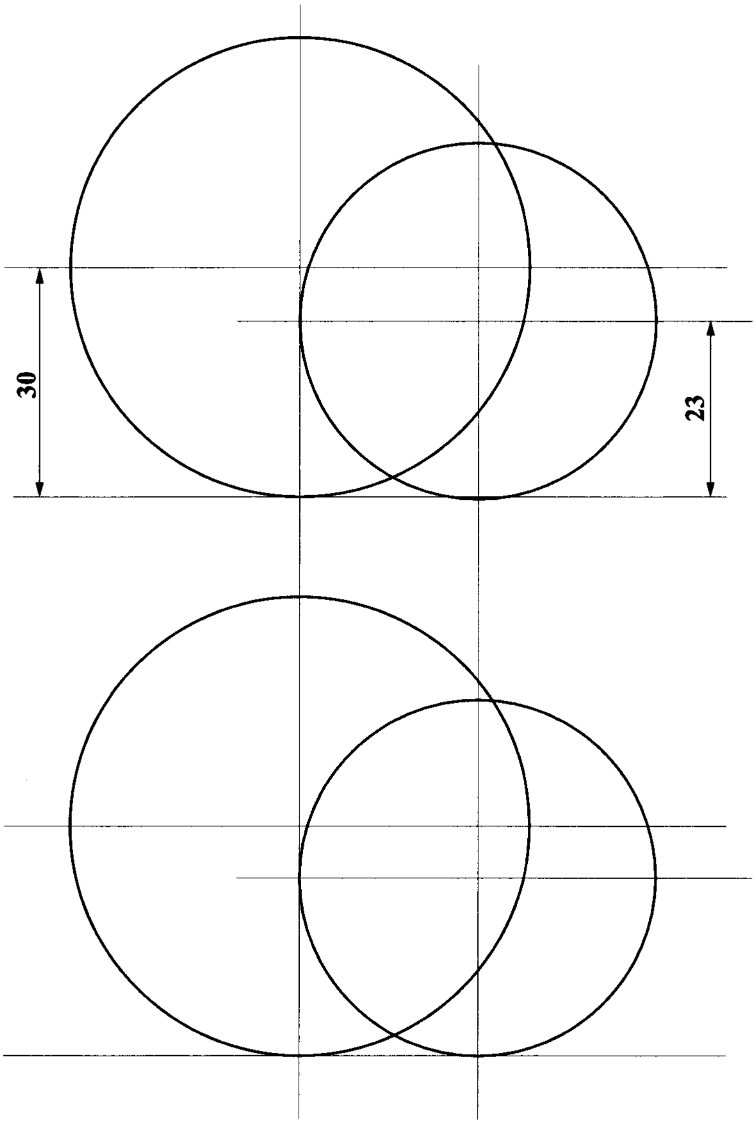

Figure 4

(a) Reproduce the given views. **[4 marks]**

(b) Complete the curve of intersection in the plan view only. **[13 marks]**

(c) Show all hidden details. **[3 marks]**

Total 20 marks

4. Figure 5 shows a component with a slanted feature.

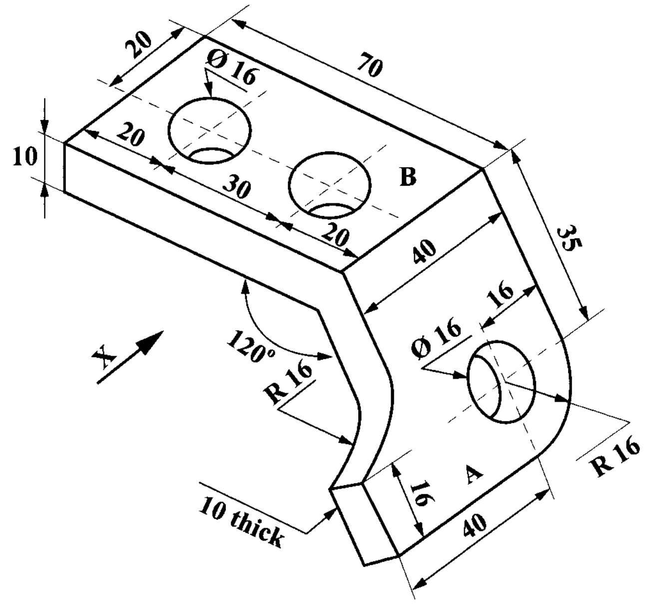

Figure 5

(a) Draw the front elevation (FE) of the component looking in the direction of arrow X. **[8 marks]**

(b) Draw the end elevation (EE) as seen from the right of the front elevation. **[6 marks]**

(c) Project an auxiliary view of the slanted feature. **[6 marks]**

Total 20 marks

NOTHING HAS BEEN OMITTED.

SECTION C

MODULE 3A — ENGINEERING DRAWING

Answer TWO questions from this section.

5. Figure 6 shows a pump and motor assembly mounted on a concrete base.

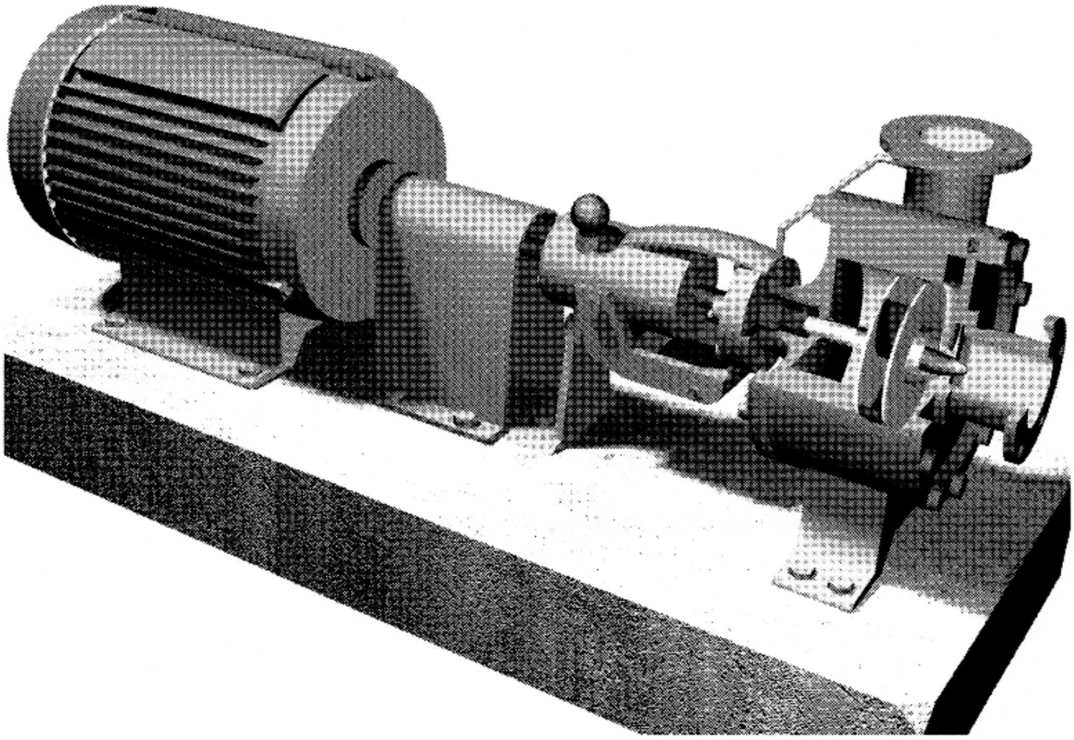

Figure 6

(a) Neatly sketch, freehand and in proportion, the assembly showing all the components clearly. **[11 marks]**

(b) Fully label EACH of the following items on your sketch:

(i) Pump inlet

(ii) Pump outlet

(iii) Coupling

(iv) Electric motor

(v) Pump impeller **[5 marks]**

(c) Name the type of pump shown in Figure 6. **[1 mark]**

(d) Explain the importance of mounting the pump and motor assembly on a concrete base. **[3 marks]**

Total 20 marks

6. Figure 7 shows the components of a C-clamp in exploded view.

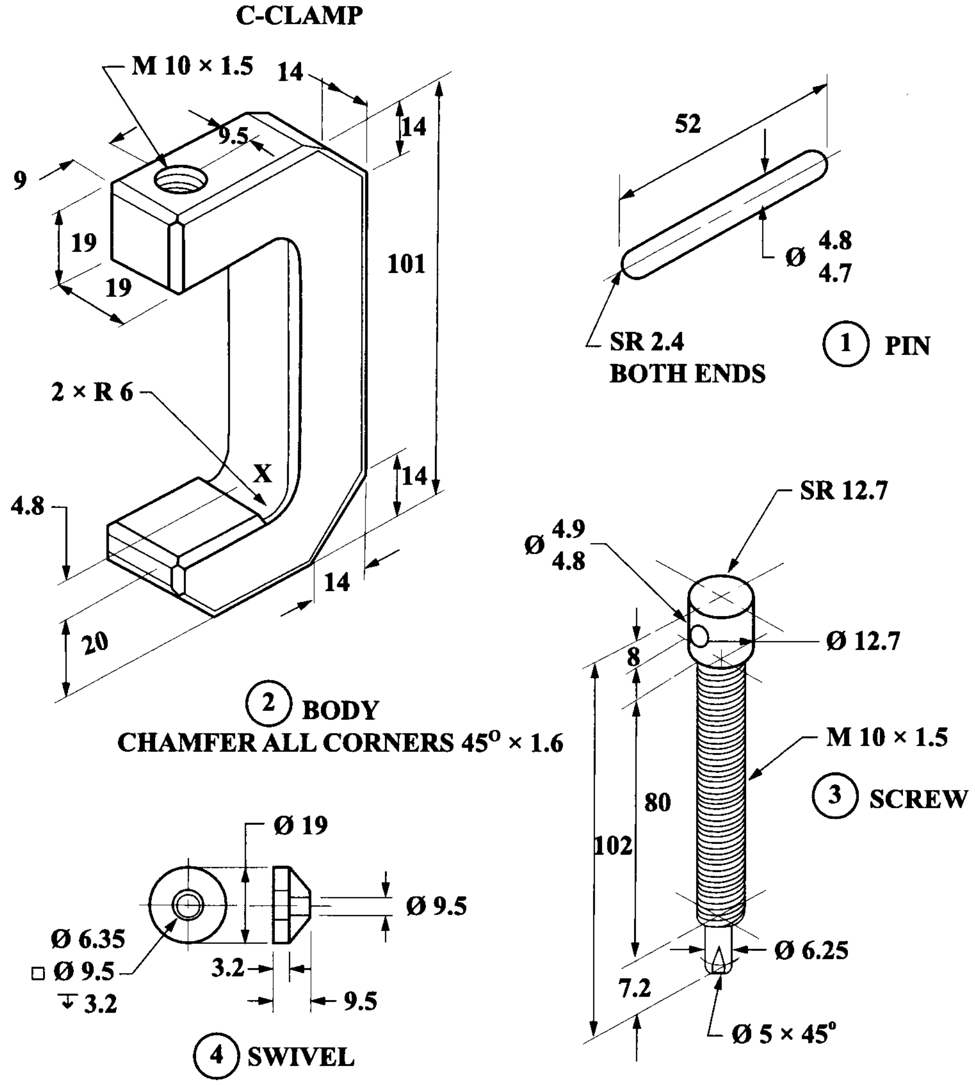

Figure 7

(a) Draw the front elevation of the C-clamp with its parts fully assembled. **[12 marks]**

(b) Use balloon referencing to identify each component of your drawing. **[4 marks]**

(c) List the components for the C-clamp in a parts list. Include the correct heading and notes. **[4 marks]**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.